MIL-C-12051B(MU)
23 June 1969
SUPERSEDING
MIL-M-12051A
29 October 1952

MILITARY SPECIFICATION

CHEMICAL AGENT, MUSTARD, DISTILLED (HD)

1. SCOPE

1.1 This specification covers one grade of vacuum distilled mustard (HD), chemically known as bis (2-chloroethyl) sulfide, $(ClCH_2CH_2)_2S$, and hereinafter referred to as HD.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-R-191 - Red Lead, Dry and Paste In Oil.
TT-W-261 - White Lead: Basic-Sulphate, Dry and Paste-In-Oil.

MILITARY

MIL-C-3250 - Cylinder, Gases and Liquids, One-Ton, Type D.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 1365

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.

DRAWINGS

US ARMY MUNITIONS COMMAND (EDGEWOOD ARSENAL)

D6-10-28 - Containers, One Ton; Marking Diagram and Bill of Materials.

PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUAL

TM 3-250 - Storage, Shipment, and Handling of Chemical Agents and Hazardous Chemicals.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.
E203-64 - Water Using Karl Fischer Reagent.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. HD shall conform to the chemical and physical characteristics of table I when tested as specified therein.

Table I. Chemical and physical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (based on freezing point)	92.0	-	4.2.4.1
Iron (as $FeCl_2$)	-	0.02	4.2.4.2
Acidity (as HCl)	-	0.10	4.2.4.3
Water content	-	0.10	4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

4.2.2 Sampling for tests of physical and chemical characteristics. Three representative specimens of no less than 1 pound each shall be randomly taken from each lot of HD prior to any filling operations. The specimens shall be placed in separate clean, dry containers labeled to identify the lot from which they were taken (see 6.4).

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. Each container shall be examined for the defects listed below. Any defect present shall be cause for rejection of the container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.

(a) Container not as specified.
(b) Container damaged.
(c) Steel tubing or threaded openings and fittings not as specified.
(d) Luting material not as specified.
(e) Marking incorrect, illegible or missing.

4.2.3.2 For test of container leakage. Each filled and closed container shall be tested for leakage as specified in 4.2.4.5. If any HD vapors are detected, the container shall be rejected.

4.2.3.3 For tests of physical and chemical characteristics. Each specimen taken in 4.2.2 shall be tested as specified in the applicable portions of 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot of HD represented.

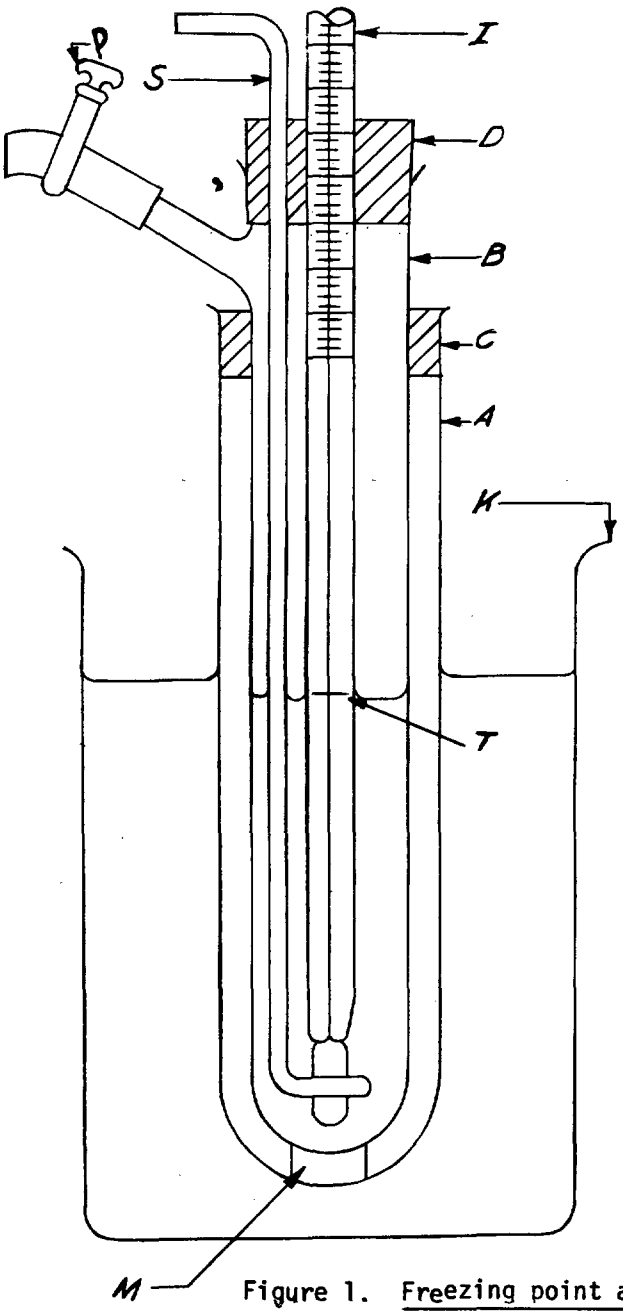
4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows (see 6.4):

4.2.4.1 Assay (based on freezing point). Using the apparatus shown on figure 1 (the thermometer should be first checked against a standardized thermometer for corrections in the appropriate temperature range), fill test tube B with specimen to the immersion mark on the thermometer. If an immersion thermometer is not used, fill the tube to 1/2 inch above the thermometer bulb. In no case should less than 25 milliliters (ml) of specimen be used. Assemble the thermometer and stirrer in a cork stopper and place in test tube B as shown on figure 1. Place test tube B in test tube A and immerse in a bath at approximately 5° C below the anticipated freezing point. Allow the temperature of the specimen to fall to 1.0° ± 0.1° C below the estimated freezing point while stirring. If crystallization does not occur before this point, it must be induced by vigorous stirring or by seeding with previously frozen crystals of the specimen. Record the highest temperature attained during crystallization as the freezing point. If a total immersion thermometer is used, correct for emergent stem. Determine percent assay by the following formula (it is assumed that for each 1 percent of impurity in the specimen, the freezing point is lowered 0.368° C). The freezing point of pure HD is taken as 14.45° C.

$$\text{Percent assay} = 100 - \frac{(14.45 - A)}{0.368}$$

where: A = Freezing point of specimen in degrees C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.



A-TEST TUBE 1 3/4" DIA. X 6 1/4" LONG.

B-TEST TUBE 1 3/8" DIA. X 7 1/4" LONG.

C-GOOCH TUBING CONNECTION.

D-STOPPER.

K-BEAKER

I-THERMOMETER GRADUATED TO READ IN TENTHS OF A DEGREE AND ARRANGED TO BE READ ABOVE STOPPER D.

M-CORK SPACER.

P-SCREW CLAMP.

S-STIRRER.

T-THERMOMETER IMMERSION MARK.

Figure 1. Freezing point apparatus

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.

Note. During freezing point determinations, HD sometimes tends to supercool, thereby making it difficult to observe the correct freezing point. In such cases, the temperature readings should be taken every 30 seconds and the temperature which remains constant (or reaches a plateau) for 1 to 2 minutes should be taken as the freezing point.

4.2.4.2 Iron. Introduce approximately 15 ml of the specimen into a dry, tared, 500-ml beaker provided with a watch glass cover and weigh the beaker and specimen to the nearest milligram (mg). Cautiously add approximately 25 ml of concentrated nitric acid dropwise from a pipet, agitating the solution frequently until the violent frothing subsides. Add 25 ml more of concentrated nitric acid and heat until the frothing ceases and the liquid is pale yellow. Cool slightly, add 20 ml of 70-percent perchloric acid, and heat with frequent agitation until dense fumes of perchloric acid are given off and the solution is light yellow. Cool and then dilute to approximately 150 ml with water. Add an excess of 6N ammonium hydroxide solution and boil for 5 minutes. Filter while warm and wash five times with hot 2-percent ammonium chloride solution. Discard the filtrate. If no brown stain of ferric hydroxide is visible on the filtering medium, the specimen contains less than 0.10 percent ferrous chloride. If a brown stain does appear, dissolve the precipitate by pouring 30 ml of hot 3N hydrochloric acid through the filter. Collect the filtrate in a 250-ml beaker or casserole. Using the Zimmermann-Reinhardt method, reduce with stannous chloride and titrate with 0.05N potassium permanganate solution. Before beginning the titration, add five drops of 0.01N sodium diphenylamine sulfonate solution as indicator. Add the potassium permanganate solution dropwise until the first tinge of purple appears and remains on stirring. Calculate the percent iron as ferrous chloride as follows:

$$\text{Percent iron} = \frac{12.7\ AB}{W}$$

where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution, and
W = Weight of specimen in grams.

4.2.4.3 Acidity. Place approximately 5 ml of the specimen in a dry, tared, 250-ml iodine flask and weigh the flask and specimen to the nearest milligram. Wash down the stopper and sides of the flask with approximately 10 ml of chloroform and then add approximately 50 ml of saturated sodium chloride solution. Stopper the flask, shake well, add ten drops of phenolphthalein indicator, and titrate with 0.05N sodium hydroxide solution. Do not shake while titrating but roll or rock the flask gently to mix the contents. The end point is defined as the point at which the pink color persists in the upper layer for a least 10 seconds upon gentle rotation of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.

flask. Run a blank test on the same quantities of chloroform and sodium chloride solution used in the specimen test. Calculate the percent acidity as hydrochloric acid as follows:

$$\text{Percent acidity} = \frac{3.65\ A\ (B - C)}{W} - 0.58\ D$$

where: A = Normality of sodium hydroxide solution,
B = Milliliters of sodium hydroxide solution used to titrate specimen,
C = Milliliters of sodium hydroxide solution used to titrate blank,
D = Percent iron calculated in 4.2.4.2, and
W = Weight of specimen in grams.

4.2.4.4 Water content. Determine percent water content in accordance with ASTM E203.

4.2.4.5 Container leakage. Remove the container to an area which is free of detectable traces of HD as determined by a blue-band detector tube from an M18A1 or M15A1 Detector Kit using the apparatus and method prescribed by the kit. Cover the ends of the container with a shield, leaving a 1-inch diameter opening for sampling the air. Test the air space in the ends for the presence of HD vapors, after a hold period of no less than 30 minutes, using the blue-band detector tube and the method and apparatus from the M18A1 or M15A1 Detector Kit.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. A quantity of 170 ± 3 gallons of HD shall be packed in a clean, dry 1-ton container conforming to MIL-C-3250. Examination, cleaning, drying, and re-examination of the container, steel tubing, and threaded openings and fittings shall be in accordance with the procedures of TM 3-250. Male threaded fittings shall be luted with basic-sulfate, white lead paste-in-oil conforming to type II of TT-W-261 or with red lead paste-in-oil conforming to type II of TT-R-191. Immediately after filling, all openings shall be securely closed and tested for leakage as specified in 4.2.4.5.

5.2 Marking. In addition to any marking required by the contract or order (see 6.2), each shipping container shall be marked as shown on US Army Munitions Command Drawing No. D6-10-28 and in accordance with Department of Transportation (DOT) regulations and MIL-STD-129. The "Poison Gas" label required by DOT regulations shall be applied to the container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. HD is intended for use as a toxic chemical agent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Additional marking, if required (see 5.2).

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.4 Sampling and testing precautions. HD in the liquid or vapor phase is poisonous and a vesicant and must be handled with extreme care. A safety directive covering the handling of HD incident to sampling and testing will be furnished the contractor upon request.

Custodian:

Army - MU

Review activity:

Army - MD

User activities:

Army - SM
Navy - OS

Preparing activity:

Army - MU(EA)

Project No. 1365-0154

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.

TO DETACH FORM - CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:15Z
Check the source to verify that this is the current version before use.